

Notes: Galle, Rodríguez-Clare, and Yi (RES, 2023)

Slicing the Pie: Quantifying the Aggregate and Distributional Effects of Trade

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1 Big picture

- **Question.** How large are the aggregate and distributional welfare effects of trade, especially the China shock, when workers are heterogeneous and sectoral reallocation is costly or imperfect?
- **Core problem.** Standard gravity trade models provide clean formulas for aggregate gains from trade, but they often hide distributional effects across worker groups. Reduced-form China-shock work documents local-labour-market losses but does not directly quantify welfare gains and losses in a structural trade model.
- **Model.** The paper builds a multi-sector Eaton–Kortum gravity model with heterogeneous workers. Workers belong to groups, such as commuting zones, and draw sector-specific abilities from a Roy–Fréchet distribution. Groups differ in comparative advantage across sectors.
- **Key parameter.** The Roy–Fréchet parameter κ governs how heterogeneous workers are across sectors. Low κ means workers are closer to sector-specific factors and distributional effects are large. High κ means workers are more easily reallocated and distributional effects are small.
- **Main theoretical result.** The model delivers a parsimonious formula for group-level welfare changes. It nests the Arkolakis–Costinot–Rodríguez-Clare aggregate gains formula but adds a sector-wage reallocation term that captures group exposure.
- **Empirical strategy.** The paper estimates κ using the relationship between China-shock-driven changes in manufacturing employment and average earnings across US commuting zones, building directly on Autor–Dorn–Hanson-style variation.
- **Estimated parameter.** The preferred estimate is around $\kappa = 1.5$, implying meaningful but not infinite sectoral specificity.
- **China shock result.** The China shock increases average US welfare, but a nontrivial number of commuting zones lose. Losses can be as high as about four times the average gain.
- **Distribution result.** The welfare effects are geographically correlated. Some regions, especially parts of Southern Appalachia and the Eastern half of the United States, have clusters of low or negative gains.
- **Inequality-adjusted welfare.** Introducing plausible inequality aversion barely reduces aggregate welfare gains. In fact, in the baseline exercise, inequality-adjusted gains are slightly larger because low-income commuting zones are not especially hurt by the China shock.
- **Extensions.** The paper extends the baseline model to include intermediate goods, imperfect substitutability between college and noncollege workers, within-commuting-zone heterogeneity and trade costs, mobility, unemployment, and labour-force participation.
- **Takeaway.** The paper “slices the pie” of gains from trade: trade produces positive average gains, but the slices are uneven. A structural trade model can quantify both aggregate gains and group-level losses using credible reduced-form variation.

2 Incremental contribution of the paper

2.1 What the paper adds relative to aggregate gains-from-trade formulas

- Arkolakis, Costinot, and Rodríguez-Clare-style results show that aggregate gains from trade can often be summarized by domestic expenditure shares and trade elasticities.
- Those formulas are powerful but aggregate by construction. They do not tell us who wins and who loses within the country.
- This paper keeps the gravity structure and the ACR intuition but adds heterogeneous worker groups and sectoral comparative advantage.

- The resulting welfare formula decomposes group welfare into an aggregate trade-price term and a group-specific sector-wage term.

Interpretation: The paper’s first incremental contribution is theoretical: it extends “new trade models, same old gains” to “same old aggregate gains, but new distributional slices.”

2.2 What the paper adds relative to reduced-form China-shock studies

- Autor, Dorn, and Hanson document large and persistent local labour market effects of Chinese import competition.
- Reduced-form estimates are informative about employment and earnings responses, but they do not directly produce welfare gains from trade.
- Galle, Rodríguez-Clare, and Yi use the ADH empirical strategy to discipline a structural model.
- The paper therefore connects credible local-labour-market variation to a welfare accounting framework.

Interpretation: The second contribution is methodological: it turns reduced-form China-shock variation into an estimate of worker-sector heterogeneity, then uses that estimate to quantify welfare.

2.3 What the paper adds relative to specific-factors trade models

- Classic specific-factors models generate clear winners and losers from trade, but they are not easily combined with modern multi-sector gravity estimation.
- The Roy–Fréchet model provides a bridge between specific factors and fully mobile labour.
- When $\kappa \rightarrow 1$, the model behaves closer to a specific-factors environment.
- When $\kappa \rightarrow \infty$, labour becomes close to perfectly mobile across sectors, and distributional effects disappear.

Interpretation: The third contribution is conceptual: the paper makes specific-factors intuition quantitatively estimable in a gravity framework.

2.4 What the paper adds through the extensions

- Intermediate goods show how input-output linkages affect welfare and compress distributional effects.
- Imperfect substitutability between college and noncollege workers shows that trade can affect skill premia and Roy gains.
- Unemployment and home production show that the baseline welfare results are robust to extensive-margin labour-market responses.
- Inequality-adjusted welfare clarifies whether trade gains remain positive after social-welfare weighting.

Interpretation: The extensions show that the baseline message is not an artifact of a narrow model: distributional effects matter, but the aggregate gains from the China shock and from trade survive several economically relevant modifications.

3 Model primitives and welfare formulas

3.1 Countries, sectors, and goods

- The world economy has countries i , sectors s , and worker groups g .

- Production and trade are governed by a multi-sector Eaton–Kortum structure.
- Countries differ in sectoral productivity, trade costs, and expenditure shares.
- Within each country, workers are heterogeneous in their sector-specific efficiency.

Interpretation: The trade side generates standard gravity structure. The labour side determines how trade shocks are distributed across worker groups.

3.2 Worker heterogeneity and Roy–Fréchet sorting

A worker from group g in country i draws sector-specific productivity from a Fréchet distribution. The parameter κ controls dispersion of comparative advantage across sectors.

- Low κ : workers are very heterogeneous across sectors, so sectoral shocks have strong distributional effects.
- High κ : workers can more easily reallocate across sectors, so sector-specific shocks are smoothed out.
- Group differences in sectoral comparative advantage appear through group-sector employment shares π_{igs} .

Interpretation: The model’s distributional content comes from the fact that different groups begin with different sectoral employment patterns and different comparative advantages.

3.3 Group-level welfare formula

A core result is that the proportional welfare change for group g in country i can be written as

$$\widehat{W}_{ig} = \widehat{P}_i^{-1} \left(\sum_s \pi_{igs} \widehat{w}_{is}^\kappa \right)^{1/\kappa}, \quad (1)$$

where $\widehat{P}_i$ is the proportional change in the consumer price index and $\widehat{w}_{is}$ is the proportional change in the sector-specific wage.

Interpretation: The first term is the aggregate consumer-price effect of trade. The second term is the group-specific wage-reallocation effect. Groups concentrated in sectors with falling wages lose more.

3.4 Connection to the ACR formula

The price-index component can be summarized using a multi-sector version of the ACR term:

$$\widehat{P}_i^{-1} \approx \prod_s \widehat{\lambda}_{iis}^{-\beta_{is}/\theta_s}, \quad (2)$$

where λ_{iis} is the domestic expenditure share, β_{is} is the sector expenditure share, and θ_s is the trade elasticity.

Interpretation: If all workers are effectively homogeneous, the group-specific wage term collapses and the standard aggregate gains-from-trade formula applies to everyone.

3.5 Aggregate welfare

With population weights ω_g , the average welfare effect is

$$\overline{W}_i = \sum_g \omega_g W_{ig}. \quad (3)$$

For proportional changes, the paper reports aggregate and average welfare effects such as

$$100(\widehat{W}_{\text{US}} - 1), \quad 100 \left(\sum_g \omega_g \widehat{W}_{\text{US},g} - 1 \right). \quad (4)$$

Interpretation: Aggregate welfare may be positive even when some groups lose. Therefore the paper separately reports mean, coefficient of variation, minimum, and maximum group-level effects.

3.6 Inequality-adjusted welfare

To incorporate inequality aversion, the paper uses an Atkinson-style social welfare function. A generic version is

$$\widehat{U}(\rho) = \left(\sum_g \omega_g \widehat{W}_g^{1-\rho} \right)^{1/(1-\rho)}, \quad (5)$$

where ρ is the coefficient of relative risk aversion or inequality aversion.

Interpretation: If poor groups lose disproportionately, inequality-adjusted gains fall. If poor groups are not the main losers, inequality-adjusted gains may be similar to or even slightly above unweighted gains.

3.7 Gains from trade

The gains from trade are computed as the negative of the proportional welfare change when moving counterfactually from the observed trade equilibrium back to autarky:

$$Gains = 100(1 - \widehat{W}^{Autarky}). \quad (6)$$

Interpretation: This asks how much welfare would fall if the US were moved back to autarky, holding the calibrated structure fixed.

4 Data and empirical implementation

4.1 Main empirical setting

- The empirical focus is the effect of the China shock on US commuting zones.
- A commuting zone is treated as a worker group g .
- Sectoral exposure comes from each commuting zone's initial employment shares across sectors.
- The China shock changes sectoral import competition and therefore sectoral wages.

Interpretation: Commuting zones are useful because they are the central unit in the ADH China-shock literature and map naturally into group-level welfare effects.

4.2 Key data sources

- Employment and earnings data follow the Autor–Dorn–Hanson construction.
- Trade data are used to measure Chinese import penetration by sector.
- Sectoral trade shares and gravity parameters are used to calibrate counterfactual trade shocks.
- The analysis uses manufacturing and non-manufacturing categories in the estimation of κ and more detailed sectors in the welfare simulations.

Interpretation: The paper combines reduced-form labour-market data with structural trade data. This allows the authors to estimate worker heterogeneity and then simulate welfare effects.

4.3 Bartik measure of import competition

The paper uses a Bartik-style import competition measure based on a commuting zone’s initial sectoral employment shares:

$$I_g = \sum_s \pi_{gs} r_s, \quad (7)$$

where r_s captures the sector-level shock from Chinese import penetration.

Interpretation: The measure says that a commuting zone is more exposed to the China shock if it was initially specialized in sectors hit harder by Chinese import competition.

4.4 Calibration of the China shock

- The China shock is calibrated as a change in China’s sectoral productivity or trade competitiveness.
- The calibration targets changes in Chinese export penetration across sectors.
- The model is then solved using exact-hat algebra, following the Dekle–Eaton–Kortum method.

Interpretation: Exact-hat algebra avoids solving for levels. It computes proportional changes in prices, wages, and welfare after a counterfactual shock.

5 Identification and estimation

5.1 Estimating κ from earnings and employment

The model implies a relationship between group-level earnings and the change in the employment share of non-manufacturing. The estimating equation is approximately

$$\ln \hat{y}_g = -\frac{1}{\kappa} \ln \hat{\pi}_{g,NM} + X_g' \Gamma + \varepsilon_g, \quad (8)$$

where $\hat{y}_g$ is the proportional change in average earnings and $\hat{\pi}_{g,NM}$ is the proportional change in the non-manufacturing employment share.

Interpretation: The slope identifies $-1/\kappa$. A more negative slope implies a smaller κ and stronger sector-specificity of workers.

5.2 Instrumental variables strategy

- The first stage predicts changes in the non-manufacturing employment share using Chinese import competition exposure.
- The instruments follow the ADH strategy, using variation in Chinese export growth to other countries or alternative Bartik measures.
- Controls include lagged manufacturing shares, Census division fixed effects, and initial demographic and labour-market characteristics.

Interpretation: The identification assumption is that the China-shock instrument affects average earnings through sectoral reallocation, not through unobserved commuting-zone shocks.

5.3 Estimation results and interpretation

- The preferred estimate of κ is around 1.5.
- Alternative specifications produce values ranging roughly from 1.4 to 2.8.
- First-stage statistics are strong in the main specifications.
- The estimates are consistent with related Roy–Fréchet estimates in labour and trade contexts.

Interpretation: A value near $\kappa = 1.5$ implies substantial sectoral heterogeneity. Workers are neither perfectly mobile nor completely sector-specific.

5.4 Threats to identification

- Commuting zones specialized in exposed sectors may have faced other contemporaneous shocks.
- Import competition could affect earnings through channels not captured by sectoral reallocation.
- The Bartik instrument may be correlated with local shocks if initial industry composition predicts unobserved trends.
- The paper addresses these issues using alternative instruments, controls, and robustness checks inspired by ADH and later Bartik-inference critiques.

Interpretation: The estimation is credible because it follows a well-known China-shock design, but the structural interpretation still depends on the model-implied mapping from reallocation to earnings.

6 Quantitative results and extensions

6.1 Aggregate and distributional effects of the China shock

- The aggregate welfare effect of the China shock is positive but small in percentage terms.
- The mean commuting-zone welfare effect is larger than the aggregate effect because the distribution is skewed.
- The coefficient of variation is high, indicating substantial heterogeneity.
- Some commuting zones experience losses; the minimum loss is around four times the average gain in the baseline.

Interpretation: The China shock is not simply good or bad. It raises average welfare while imposing geographically concentrated losses.

6.2 Geographic distribution

- Low or negative welfare effects are spatially clustered.
- Southern Appalachia and parts of the Eastern United States are especially exposed.
- Coastal and less manufacturing-intensive regions often fare better.

Interpretation: The model gives a welfare interpretation to the geographical heterogeneity emphasized in the China-shock literature.

6.3 Inequality-adjusted welfare

- Inequality-adjusted welfare effects are close to the average welfare gains.
- Plausible values of inequality aversion do not overturn the positive aggregate welfare effect.
- This occurs because low-income commuting zones are not systematically the biggest losers from the China shock.

Interpretation: Distributional losses are important, but they are not concentrated enough among poor groups to eliminate aggregate gains under standard inequality aversion.

6.4 Gains from trade

- The aggregate gains from all trade are larger than the gains from the China shock.
- For $\kappa = 1.5$, aggregate gains from trade are about 1.56%.
- Some commuting zones lose from trade in the model, but the majority gain.
- Lower κ implies larger dispersion and stronger distributional effects.

Interpretation: The model preserves positive aggregate gains from trade while allowing meaningful local losses.

6.5 Intermediate goods

- Allowing for input-output linkages raises aggregate gains because imported intermediate goods reduce production costs.
- It also compresses some distributional effects because sectors are linked through production networks.
- The China shock and gains from trade are both larger than in the baseline model.

Interpretation: Intermediate goods strengthen aggregate gains and somewhat mitigate dispersion, in line with input-output gains-from-trade logic.

6.6 College and noncollege workers

- The extension treats college and noncollege workers as imperfect substitutes.
- It allows trade shocks to affect the college premium and group-level Roy gains.
- The results are very similar to the baseline, with modest changes in the college premium.

Interpretation: Skill heterogeneity matters, but the main aggregate and distributional conclusions are robust.

6.7 Unemployment and home production

- The final extension allows for search-and-matching unemployment and labour-force participation.
- The model is estimated by GMM using equations for income, unemployment, and non-manufacturing employment.
- Search frictions can amplify welfare effects, but home production dampens them because people can exit employment into home production.
- Quantitative results remain close to the baseline.

Interpretation: The welfare effects are robust to labour-market margins beyond sectoral reallocation.

7 Tables and figures

7.1 Table 1: Estimation of κ

Read:

- The dependent variable is the log change in average earnings.
- The key explanatory variable is the log change in the non-manufacturing employment share.
- The coefficient is negative, consistent with the model’s prediction that movement out of manufacturing is associated with earnings changes.
- The implied κ values range from about 1.42 to 2.79 across specifications.
- The preferred specification gives $\kappa \approx 1.56$, and the paper adopts $\kappa = 1.5$ for quantitative analysis.

Economic message: The table converts reduced-form China-shock variation into the key structural parameter governing worker reallocation and distributional effects.

	(1) $\ln \hat{y}_t$	(2) $\ln \hat{y}_t$	(3) $\ln \hat{y}_t$	(4) $\ln \hat{y}_t$
$\ln \hat{\pi}_{NM}$	-0.358* (0.211)	-0.639** (0.303)	-0.704** (0.295)	-0.487** (0.183)
Implied κ	2.79 (1.643)	1.56 (0.742)	1.42 (0.594)	2.05 (0.773)
F-first stage	58.46	24.02	29.52	67.87
Observations	1,444	1,444	1,444	1,444
R^2	0.683	0.667	0.662	0.677
Import penetration	Other (lagged)	Other (no lag)	US (no lag)	US Bartik

Notes: IV-estimation results for specification (23), where y_t is average earnings per worker, and π_{NM} is the employment share in non-manufacturing, measured using the CBP data. The columns differ in the construction of the instrument: column (1) uses the exact instrument borrowed from ADH $Z_{gt} \equiv \sum_{i \in M} \pi_{gst-10} \Delta IP_{gt}^{China \rightarrow Other}$, column (2) uses $Z_{gt} \equiv \sum_{i \in M} \pi_{gst} \Delta IP_{gt}^{China \rightarrow Other}$, column (3) uses $Z_{gt} \equiv \sum_{i \in M} \pi_{gst} \Delta IP_{gt}^{China \rightarrow US}$, and column (4) uses our Bartik variable for the US: $Z_{gt} \equiv \ln \sum_i \pi_{gst} \hat{r}_{gt}$. Due to data constraints on π_{gst-10} , we have not constructed $Z_{gt} \equiv \sum_{i \in M} \pi_{gst-10} \Delta IP_{gt}^{China \rightarrow US}$. Standard errors are clustered at the state level and reported in parentheses, with * $p < 0.1$, ** $p < 0.05$, *** $p < 0.001$. The first row shows the second-stage results, while the third row has the corresponding κ estimates implied by the model and the fifth row displays the F-statistic from the first stage. All regressions include the same controls employed in ADH’s preferred specification: lagged manufacturing shares, period fixed effects, Census division fixed effects, and beginning-of-period conditions (% college educated, % foreign-born, % employment among women, % employment in routine occupations, and the average offshorability index).

7.2 Table 2 and Figure 1: Welfare effects from the rise of China

Read:

- Table 2 shows that the aggregate welfare gain from the China shock is about 0.22% for $\kappa = 1.5$.
- The mean commuting-zone gain is about 0.29%, but the coefficient of variation is 1.01.
- The minimum effect is -1.18% and the maximum is 1.65%, showing large dispersion.
- Figure 1 maps the commuting-zone welfare effects and shows strong geographic clustering.
- Some of the worst-hit areas are concentrated in manufacturing-intensive regions.

Economic message: The China shock raises average welfare but creates substantial local losses. This is the core “slicing the pie” result.

TABLE 2
The welfare effects of the China shock on the US

κ	$\hat{W}_{US}$	Mean	CV	Min.	Max.	$\prod_s \hat{\lambda}_s^{-\beta_s/\theta_s}$
$\rightarrow 1$	0.24	0.32	1.24	-1.44	2.30	0.14
1.5	0.22	0.29	1.01	-1.18	1.65	0.15
	(0.02)	(0.03)	(0.23)	(0.3)	(0.6)	(0.01)
3.0	0.20	0.26	0.70	-0.74	0.96	0.16
$\rightarrow \infty$	0.20	0.20	0	0.20	0.20	0.20

Notes: The first column displays the aggregate welfare effect of the China shock for the US, in percentage terms $100(\hat{W}_{US} - 1)$, and the second column shows the mean welfare effect: $100(\frac{1}{G} \sum_g \hat{W}_{US,g} - 1)$. The third column shows the coefficient of variation (CV), and for the fourth and fifth column, we have $\text{Min.} \equiv \min_g 100(\hat{W}_{US,g} - 1)$ and $\text{Max.} \equiv \max_g 100(\hat{W}_{US,g} - 1)$, respectively. The final column displays the multi-sector ACR term $100(\prod_s \hat{\lambda}_{US,s}^{-\beta_s/\theta_s} - 1)$. The values for $\hat{W}_{US,g}$ are calibrated for $\kappa = 1.5$. The third row has standard errors in parentheses, computed using the delta method and numerical derivatives with respect to $\hat{\beta} = 1/\hat{\kappa}$, for each statistic when $\kappa = 1.5$. We provide robustness checks for these numbers in [Supplementary Appendix Table F.3](#).

Table 2: welfare effects from the rise of China

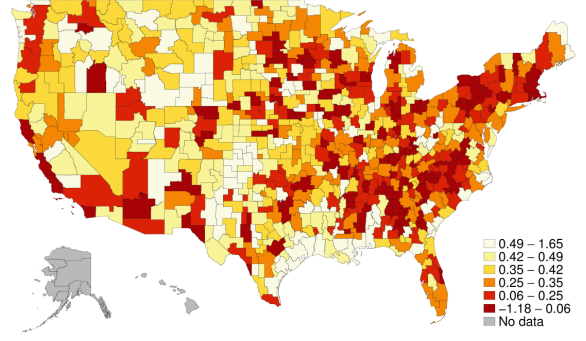


FIGURE 1
Geographical distribution of the welfare effects from the rise of China
Notes: This figure plots the geographic distribution of $100(\hat{W}_g - 1)$, where $\hat{W}_g$ are the welfare effects for group g in the US from the counterfactual rise of China, for our preferred value of $\kappa = 1.5$.

Figure 1: geography of welfare effects

7.3 Figure 2: Inequality-adjusted welfare and income exposure

Read:

- Panel A plots inequality-adjusted welfare gains as inequality aversion ρ rises.
- For plausible ρ values, the inequality-adjusted gain is close to the average gain.
- Panel B shows that the relationship between initial income and import-competition-induced income change is mildly positive.
- Low-income commuting zones are not disproportionately exposed to the worst welfare effects.

Economic message: Distributional losses matter, but they are not strongly targeted at poorer commuting zones. Hence inequality aversion barely reduces the welfare gains.

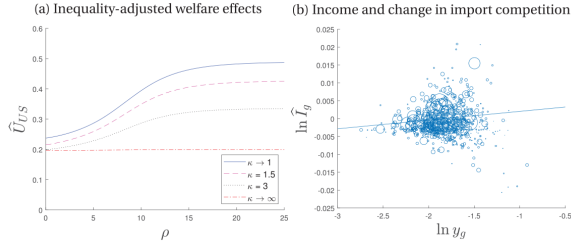


FIGURE 2
Inequality-adjusted welfare effects of the China shock

Notes: (a) The relationship between the inequality-adjusted welfare effects of the rise of China $\hat{U}_{US} \equiv (\sum_g w_g \hat{W}_{US,g}^{1-\rho})^{\frac{1}{1-\rho}}$ and ρ . Here, ρ is the coefficient of relative risk aversion for the agent behind the veil of ignorance and $w_g \equiv \frac{y_g(V_g/L_g)^{1-\rho}}{\sum_h y_h(V_h/L_h)^{1-\rho}}$ a modified weight for group g . The vertical axis displays $100(\hat{U}_{US} - 1)$. (b) The relationship between $\ln \hat{I}_g = \ln(\sum_g \pi_{gR} \hat{r}_{gR})^{-1}$, our measure for the change in regional import-competition (computed for $\kappa = 1.5$), and the logarithm of a group's average income per worker. The solid line displays the linear fit for this relationship, with each commuting zone weighted by its population size. The size of a circle indicates the population size for that commuting zone.

TABLE 3
Aggregate and Group-level Gains from Trade

κ	$\hat{W}_{US}$	Mean	CV	Min.	Max.	$\prod_s \hat{\lambda}_s^{-\beta_s/\theta_s}$
$\rightarrow 1$	1.61	1.61	0.90	-7.95	3.81	1.45
1.5	1.56	1.56	0.63	-4.82	3.03	1.45
	(0.05)	(0.05)	(0.26)	(2.85)	(0.71)	(0)
3.0	1.50	1.50	0.33	-1.68	2.24	1.45
$\rightarrow \infty$	1.45	1.45	0	1.45	1.45	1.45

Notes: The first column displays the aggregate gains from trade for the US, in percentage terms $(100(1 - \hat{W}_{US}))$ and the second column shows the mean welfare effect: $100(\frac{1}{G} \sum_g 1 - \hat{W}_{US,g})$. Here, $\hat{W}_{US}$ and $\hat{W}_{US,g}$ are the aggregate and group-level welfare change from a return to autarky for the US. The third column shows the coefficient of variation (CV), and for the fourth and fifth column we have $\text{Min.} \equiv \min_g 100(1 - \hat{W}_{US,g})$ and $\text{Max.} \equiv \max_g 100(1 - \hat{W}_{US,g})$, respectively. The final column displays the multi-sector ACR term $100(1 - \prod_s \hat{\lambda}_{US,s}^{-\beta_s/\theta_s})$. The third row has standard errors in parentheses, computed using the delta method and numerical derivatives with respect to $\hat{\beta} = 1/\hat{\kappa}$, for each statistic when $\kappa = 1.5$. [Supplementary Appendix Table F.3](#) provides a robustness check with sector-specific θ_s values.

7.4 Table 3: Aggregate and group-level gains from trade

Read:

- For $\kappa = 1.5$, aggregate gains from trade are about 1.56%.
- The coefficient of variation is 0.63, indicating distributional heterogeneity but less than in the China-shock exercise.
- The minimum effect is -4.82% and the maximum is 3.03%.

- As κ increases, distributional dispersion falls.

Economic message: Trade delivers positive aggregate gains, but the gains are not evenly distributed across commuting zones.

7.5 Table 4: Intermediate goods

Read:

- Adding input-output linkages raises aggregate gains.
- For the China shock, the aggregate effect rises from the baseline 0.22% to 0.37%.
- Gains from trade rise to about 2.86%.
- Dispersion is somewhat compressed relative to the baseline.

Economic message: Input-output linkages amplify aggregate gains because imported inputs reduce production costs, but they also spread effects through production networks.

TABLE 4
Counterfactual analysis for the model with intermediates

	$\hat{W}_{US}$	Mean	CV	Min.	Max.	ACR
Rise of China	0.37 (0.02)	0.45 (0.04)	0.54 (0.14)	-0.90 (0.35)	1.28 (0.42)	0.29 (0.01)
Gains from trade	2.86 (0.05)	2.92 (0.08)	0.29 (0.12)	-2.12 (2.2)	4.12 (0.62)	2.74 (0)

Notes: The tables show summary statistics for welfare effects of US groups for the model with an input-output structure for $\kappa = 1.5$. The first two rows show results for the counterfactual rise of China, and the final two rows show results for group-level gains from trade. The first column displays the aggregate welfare effect for the US, in percentage terms $100(\hat{W}_{US} - 1)$ and the second column shows the mean welfare effect: $100(\frac{1}{G} \sum_g \hat{W}_{US,g} - 1)$. The third column shows the coefficient of variation (CV), and for the fourth and fifth column we have $\text{Min.} = \min_g 100(\hat{W}_{US,g} - 1)$ and $\text{Max.} = \max_g 100(\hat{W}_{US,g} - 1)$, respectively. The final column displays the multi-sector ACR term $100(\prod_{i,k} \hat{w}_{US,i,k}^{\beta_{i,k}} - 1)$. For the gains from trade, we simulate the return to autarky and report the negative of the above statistics for the obtained

TABLE 5
College and non-college workers as imperfect substitutes

(a) The rise of China

	$\hat{W}_{US}$	Mean	CV	Roy gains	College premium
$\kappa \rightarrow \infty; \eta = 1.6$	0.15	0.15	0.15	0.00	-0.03
$\kappa = 1.5; \eta = 1.6$	0.22 (0.02)	0.31 (0.04)	0.77 (0.17)	0.07 (0.03)	-0.01 (0.02)
$\kappa = 1.5; \eta \rightarrow \infty$	0.22	0.32	0.73	0.07	0.00

(b) Gains from trade

	$\hat{W}_{US}$	Mean	CV	Roy gains	College premium
$\kappa \rightarrow \infty; \eta = 1.6$	1.45	1.48	0.14	0.00	1.00
$\kappa = 1.5; \eta = 1.6$	1.56 (0.05)	1.60 (0.06)	0.47 (0.18)	0.11 (0.05)	0.10 (0.09)
$\kappa = 1.5; \eta \rightarrow \infty$	1.56	1.49	0.59	0.11	0.00

Notes: The table presents the welfare effects for trade shocks for the model with college and non-college workers as potentially imperfect substitutes. Panel (a) shows results for the rise of China and panel (b) for the gains from trade. The first column provides the parameter values for the simulation results in that row, where $\eta \rightarrow \infty$ implies that all labour is perfectly substitutable. Column 2 displays the aggregate welfare effect for all workers in the US, in percentage terms $100(\hat{W}_{US,m} - 1)$, column 3 shows the mean welfare effect: $100(\frac{1}{G} \sum_g \hat{W}_{US,m,g} - 1)$, and column 4 the coefficient of variation (CV). The fifth column shows the aggregate Roy gains $100(\sum_{m \in \mathcal{M}} (\frac{Y_m}{Y_i}) \prod_j \hat{z}_{m,j}^{\beta_{m,j}/\kappa} - 1)$, and the final column the change in the college premium $100(\prod_i (\hat{z}_{iC}/\hat{z}_{iNC})^{-\beta_{iC}/(\mu-1)} - 1)$. The China shock is separately calibrated for the parameter values in each row. For the gains from trade, we simulate the return to autarky, and for that simulation report the negative of the aggregate and mean welfare effect. Standard errors for the benchmark results in the second row, computed using the delta method

7.6 Table 5: College and noncollege workers

Read:

- The table compares perfect substitutability with imperfect substitutability between college and noncollege workers.
- The China-shock aggregate gain remains about 0.22% in the preferred imperfect-substitutes specification.
- Gains from trade remain about 1.56%.
- Changes in the college premium are small.

Economic message: Skill heterogeneity and imperfect substitution do not overturn the baseline welfare conclusions.

7.7 Table 6 and Table 7: Employment, unemployment, and home production

Read:

- Table 6 estimates the model with unemployment and labour-force participation using GMM.
- The preferred calibration uses $\kappa = 1.5$, $\alpha = 1/3$, and $\mu = 2.5$.

- Table 7 compares welfare effects with and without search-and-matching unemployment and home production.
- Search and matching amplifies some effects, while home production dampens them.
- Overall welfare effects remain close to the baseline.

Economic message: The baseline conclusions are robust to adding realistic labour-market margins.

TABLE 6
GMM Estimation of the Model with Unemployment and Labour Force Participation

	(1)	(2)	(3)	(4)
$\ln \hat{E}$	0.985* (0.563)	2.398* (1.456)	4.389 (3.199)	1.002** (0.408)
$\ln \hat{\pi}_{HP}$	-0.312** (0.122)	-0.519 (0.369)	-0.760 (0.708)	-0.362*** (0.104)
$\ln \hat{\pi}_{NM}$	-0.633** (0.291)	-0.972** (0.409)	-1.001** (0.409)	-0.834** (0.258)
Implied α	0.504	0.294	0.186	0.499
Implied μ	3.209	1.927	1.316	2.759
Implied κ	1.579	1.029	0.999	1.199
Observations	1,444	1,444	1,444	1,444
Import penetration	Other (lagged)	Other (no lag)	US (no lag)	US Bartik

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Standard errors are clustered at the state level, and we weight by 1990 commuting

Table 6: GMM estimation of employment model

TABLE 7
Welfare effects with and without frictional unemployment and home production

	The rise of China			Gains from trade		
	$\hat{W}_{US}$	Mean	CV	$\hat{W}_{US}$	Mean	CV
No SAM, no HP	0.2150	0.294	1.010	1.556	1.556	0.626
With SAM, no HP	0.3240	0.443	1.001	2.325	2.327	0.616
With SAM & HP, $\mu = 1$	0.2116	0.288	1.022	1.531	1.529	0.626
With SAM & HP, $\mu = 2.5$	0.2121	0.288	1.011	1.521	1.520	0.618

Notes: The first three columns display the welfare effects for the counterfactual rise of China, while the final three columns show the gains from trade. Columns 1 and 4 displays, for the relevant worker type, the aggregate welfare effect in percentage terms $100(\hat{W}_{US} - 1)$, and columns 2 and 5 show the mean welfare effect: $100(\frac{1}{L} \sum_g \hat{W}_{US,g} - 1)$. The third column shows the coefficient of variation (CV). For the gains from trade, we simulate the return to autarky, and for that simulation report the negative of the above statistics. Full results are available in [Supplementary Appendix Tables 1.5 and 1.6](#). We compute standard errors for these counterfactual exercises in [Supplementary Appendix Table 1.7](#).

Table 7: welfare effects with unemployment and home production

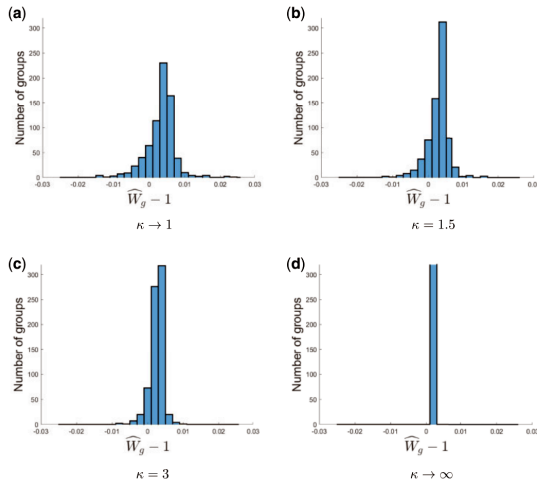
7.8 Appendix diagnostics and robustness visuals

Read:

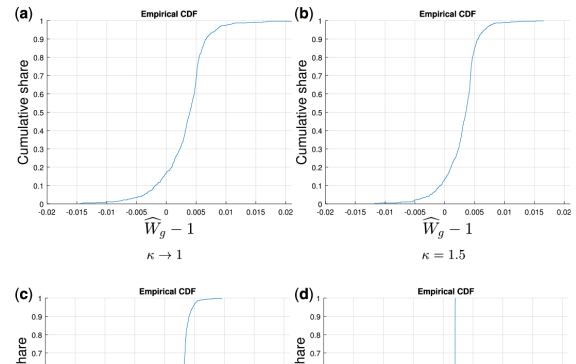
- Appendix histograms and CDFs show how the distribution of welfare effects changes with κ .
- As κ rises, distributional dispersion shrinks.
- Appendix import-competition figures illustrate the relationship between Bartik exposure and income changes.
- Appendix tables provide sector exposure and robustness regressions.

Economic message: The appendix diagnostics reinforce the main mechanism: worker heterogeneity governs the dispersion of gains and losses.

Appendix A. Supplementary Tables and Figures for the Counterfactuals



Appendix Figure A.1: distribution of China-shock effects



Appendix Figure A.2: CDFs by κ

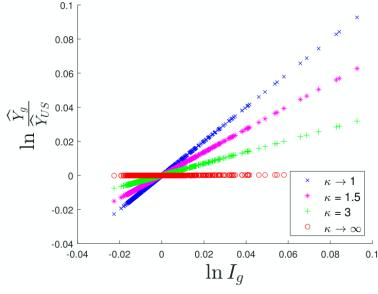
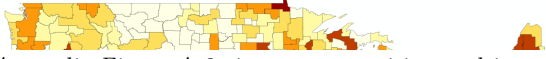


FIGURE A.8

Import competition and groups' relative gains from return to autarky

Notes: The figure plots the value for $\ln \frac{Y_g^g}{Y_g^S}$ in relation to $\ln I_g = \sum_i \pi_{gi} \frac{\beta_{gi}}{r_i}$, our Bartik measure for groups' import-competition. Each scatter represents the simulation results for the return to autarky for a different value of κ .



Appendix Figure A.8: import competition and income

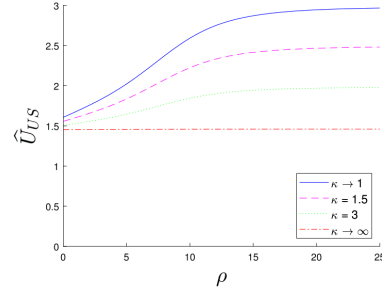


FIGURE A.10

Inequality-adjusted gains from trade

Notes: The figure plots the relationship between the inequality-adjusted gains from trade $\hat{U}_{US} = \left(\sum_g \omega_g \hat{W}_g^{1-\rho} \right)^{\frac{1}{1-\rho}}$ and ρ . Here, ρ is the coefficient of relative risk aversion for the agent behind the veil of ignorance and $\omega_g = \frac{I_g(Y_g/I_g)^{1-\rho}}{\sum_h (I_h/I_h)^{1-\rho}}$ a modified weight for group g . The vertical axis displays $100(1 - \hat{U}_{US})$.

TABLE A.1
List of sectors

Sector Nr.	Sector description	β_g	r_g	β_g/r_g	$\lambda_{US,US,g}$
15–16	Food, Beverages, and Tobacco	0.03	0.03	0.98	0.95
17–19	Textiles and Textile of Leather Products	0.01	0.01	1.52	0.57
20	Wood and Products of Wood and Cork	0.01	0.01	1.09	0.86
21–22	Pulp, Paper, Printing, and Publishing	0.02	0.02	0.98	0.94
23	Coke, Refined Petroleum, and Nuclear Fuel	0.01	0.01	1.03	0.91
24	Chemicals and Chemical Products	0.02	0.02	1.01	0.82
25	Rubber and Plastics	0.01	0.01	1.01	0.89
26	Other Non-Metallic Mineral	0.01	0.01	1.06	0.85
27–28	Basic Metals and Fabricated Metal	0.03	0.02	1.06	0.86
29	Machinery, Nec	0.02	0.02	0.95	0.76
30–33	Electrical and Optical Equipment	0.04	0.04	1.07	0.63
34–35	Transport Equipment	0.04	0.03	1.06	0.73
36–37	Manufacturing, Nec; Recycling	0.01	0.01	1.26	0.67
	Non-manufacturing	0.75	0.76	0.98	0.98

Notes: This table lists the 14 sectors used in our analysis. The first column has the ISIC Rev.3 sectors for each of the manufacturing subsectors, and the second column has the sector description. The next three columns show the Cobb-Douglas expenditure share, the earnings share r_g and the sectoral import-competition index β_g/r_g for the US. The final column has the domestic expenditure share for the US, $\lambda_{US,US,g}$.

TABLE A.2
Fit of China shock counterfactuals to the data

	(1) Actual $\ln \hat{e}_g$	(2) Actual $\ln \hat{\pi}_{gHP}$	(3) Actual $\ln \hat{e}_g$	(4) Actual $\ln \hat{\pi}_{gHP}$
Model-predicted $\ln \hat{e}_g$ ($\alpha = 1/3$)	1.324 (1.734)			
Model-predicted $\ln \hat{\pi}_{gHP}$ ($\alpha = 1/3$)		2.047*** (0.562)		
Model-predicted $\ln \hat{e}_g$ ($\alpha = 1/2$)			1.925**	

Appendix Table A.1: sectors and exposure

Appendix figure: gains-from-trade diagnostics

TABLE A.3
Regression of predicted changes in employment and income per worker on ADH's China shock

	(1) $\ln \hat{\pi}_{gHP}$	(2) $\ln \hat{e}_g$	(3) $\ln \hat{y}_g$
$\sum_{i \in M} \pi_{gi} \Delta MP_{gi}^{China \rightarrow USA}$	0.284*** (0.0834)	-0.0379*** (0.0111)	-0.0758*** (0.0222)
Mean and SD of dependent variable	-0.557 [0.0566]	0.0742 [0.0754]	0.148 [0.151]
First stage coeff.	0.528	0.528	0.528
First stage F	29.30	29.30	29.30
Observations	722	722	722
Instrument	China $\rightarrow$ Other	China $\rightarrow$ Other	China $\rightarrow$ Other

Notes: *** $p < 0.01$. Standard errors, in parentheses, are clustered at the state level. The table regresses model predicted changes in employment and income per worker on the actual ADH's China shock for the period 2000–7. All dependent variables are based on Section 8's model and assume $\alpha = 1/3$ (for ease of exposition, all dependent variables are multiplied by 100). We also provide the means and standard deviations of the dependent variables in rows 3 and 4). The regressor of interest is the original ADH import exposure measure ($\sum_{i \in M} \pi_{gi} \Delta MP_{gi}^{China \rightarrow USA}$), which is instrumented using the ADH instrumental variable: $\sum_{i \in M} \pi_{gi} \Delta MP_{gi}^{China \rightarrow Other}$. All regressions include the same controls employed in ADH's preferred specification: lagged manufacturing shares, period fixed effects, Census division fixed effects, and beginning-of-period conditions (% college educated, % foreign-born, % employment among women, % employment in routine occupations, and the average offshorability index).

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Appendix Table A.2: employment and income regressions

8 Short summary

- The paper develops a multi-sector gravity model with heterogeneous workers and group-specific comparative advantage.
- It nests the ACR aggregate gains formula but adds a group-level reallocation term.
- The key worker-heterogeneity parameter κ is estimated using China-shock variation across US commuting zones.
- The preferred estimate is around $\kappa = 1.5$.
- The China shock generates positive average welfare gains but also losses for a subset of commuting zones.
- Losses can be several times the average gain.

- Inequality-adjusted gains remain positive and close to the average gains because poor commuting zones are not disproportionately hurt.
- Gains from overall trade are larger than gains from the China shock, but still unevenly distributed.
- Extensions with intermediate goods, skill groups, and unemployment preserve the main qualitative results.
- The paper provides a structural bridge between the China-shock reduced-form literature and aggregate gains-from-trade theory.

9 Conclusion

9.1 Core conclusion

Galle, Rodríguez-Clare, and Yi show that trade can raise aggregate welfare while creating meaningful local losses. Their model keeps the tractability of gravity trade frameworks and the ACR aggregate-gains logic, but adds heterogeneous workers and group-level sectoral exposure. This lets the authors quantify both the average gain and the distribution of gains and losses.

9.2 Economic intuition

The intuition is that trade shocks change sectoral wages and prices. Groups initially concentrated in sectors hurt by import competition lose more, especially when workers have sector-specific comparative advantage and cannot easily reallocate. A low κ makes workers more like specific factors, while a high κ makes them more mobile across sectors. The estimated value near 1.5 implies meaningful but not maximal distributional effects.

9.3 Takeaway

The paper’s main lesson is that the welfare case for trade should not be stated only in aggregate terms. Aggregate gains are positive, but the distribution of gains matters. A credible welfare accounting must identify who gains, who loses, by how much, and how those losses map across regions and income groups. This paper provides one of the clearest structural frameworks for doing exactly that.